

ManOSalwaKnot

AI-assisted food rescue and surplus-food redistribution platform for Pakistan

v0.1 · Draft for review

4 October 2026

Pickup-first MVP

Delivery in V2

All claims are hypotheses to be validated.

MVP SUCCESS LOOP

Surplus

Listing

Moderated

Discover

Reserve

Pay

Pickup

Measured

Owner: Product Owner (name not provided, OQ-01) · Source: input draft PRD (47 sections), consolidated

Stack: React/TypeScript/Vite/Tailwind · FastAPI · Supabase · Redis · LangChain/LangGraph · Groq · Gemini · Tavily · Resend

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How to read this document. Strength convention: **must** = mandatory, **should** = expected unless justified, **may** = optional. Open items are tagged **[OQ-nn]** (in red) and listed in Section 23. Items marked *proposed* are not yet confirmed and need Product Owner review.

1 Document Control

Product	Version	Status	Date	Owner	Source
ManOSalwaKnot	0.1 (concise edition)	Draft for review	2026-10-04	Product Owner, name not provided [OQ-01]	Input draft PRD (47 sections), consolidated

Change log. v0.1 (2026-10-04): restructured the input draft into 24 sections, merged duplicates, added acceptance criteria and open questions.

Stakeholders (names not provided): Product Owner, Engineering, pilot Providers, pilot Recipients/NGOs, Admin/Moderators, Investors, Compliance/food-safety reviewer.

2 Executive Summary

ManOSalwaKnot is a Pakistan-wide, AI-assisted food rescue and surplus-food redistribution platform. It links food businesses, institutions, NGOs and individuals through trusted, hyperlocal coordination. Providers publish surplus, set pickup windows and manage reservations. Recipients discover food by budget, distance and quantity, reserve it, pay by cash or online where available, and use the AI assistant **Salwa**.

It is **not** a food-delivery app. The MVP is pickup-first; delivery is deferred to V2 via third-party partners. **AI assists; rules and humans decide; AI must never certify food as safe from an image alone.** The business model is B2B-first and hybrid, and every monetization and adoption claim is a hypothesis to be validated.

MVP success loop: Provider has surplus → listing created → moderated → recipient discovers → reserves → pays → picks up → transaction complete → waste reduction measurable.

Priority: reliable food rescue first; advanced AI, delivery and monetization second.

Baseline stack: React / TypeScript / Vite / Tailwind · FastAPI · Supabase (PostgreSQL, Auth, RLS) · Redis · LangChain / LangGraph, Groq LLM, Gemini Vision, Tavily · Resend · Vercel, Render.

3 Problem Statement & Opportunity

Providers hold edible surplus from unpredictable demand, end-of-day stock, events, overproduction, cancelled orders and short shelf life. Existing alternatives fail on seven points:

ID	Problem	Description
P-1	Discovery	No view of nearby surplus
P-2	Speed	Short useful window, no fast liquidation
P-3	Trust	Provider identity, listing and hygiene doubts
P-4	Coordination	No structured pickup or reservation; checkout and payment friction
P-5	Affordability	Food cost is a barrier for low-budget recipients
P-6	Visibility	No operational insight for providers
P-7	Waste tracking	Impact is not measurable

Opportunity: a digital layer with hyperlocal discovery, controlled reservation and pickup, a trust layer and AI assistance. Market size is not quantified in the source **[OQ-02]**.

4 Vision, Mission & Product Positioning

Vision: make surplus food easy to rescue, trusted to discover, and simple to redistribute.

Mission: reduce avoidable food waste in Pakistan through trusted digital infrastructure that moves surplus quickly to people and organizations that can use it.

Formula: Food Rescue + Marketplace + AI Coordination + Trust Layer

User positioning: “Find good surplus food near you, reserve it, and pick it up before it goes to waste.”

Investor positioning: digital infrastructure for surplus-food redistribution in Pakistan, combining a trusted marketplace, pickup coordination, AI matching and business intelligence.

Priority hierarchy (top wins)

1 Food Rescue → 2 Trusted Marketplace → 3 Pickup & Reservation → 4 AI Assistance → 5 Business Intelligence → 6 Delivery & Advanced Monetization

Product principles

ID	Principle	Requirement
PR-1	Trust first	Show provider, food, quantity, price/free, pickup window, handling info and status.
PR-2	Pickup first	MVP uses controlled pickup only.
PR-3	AI assists, humans/rules decide	AI must not certify food as safe.
PR-4	Fast listing	Listing creation in minimal steps.
PR-5	Hyperlocal	Prioritize nearby, time-sensitive surplus.
PR-6	Privacy by design	No unnecessary personal data or private addresses to the public.
PR-7	Mission plus business	Reduce waste; B2B revenue is a hypothesis.

5 Goals, Non-Goals & Success Metrics

Goals

- **G-01** prove the full rescue loop in a controlled pilot
- **G-02** fast listing by verified providers
- **G-03** easy discover → reserve → collect
- **G-04** trust layer (verification, moderation, incidents, audit)
- **G-05** AI makes the loop faster without adding complexity
- **G-06** measurable waste reduction
- **G-07** gather evidence on business hypotheses

Non-goals (must not become)

- **NG-01** generic delivery marketplace · **NG-02** unsupervised AI food-safety authority · **NG-03** personal-data marketplace · **NG-04** rider-fleet company at MVP
- **NG-05** social network · **NG-06** anonymous public food marketplace · **NG-07** platform charging users merely for access to donated food **[OQ-03]** · **NG-08** AI-first product with a weak core workflow

All KPI targets are **TBD** and must be set before the pilot **[OQ-04]**.

KPI definitions

Group	ID	Definition and measurement
Mission	M-01	Servings redistributed: quantity on Collected reservations, converted to servings [OQ-05]

Group	ID	Definition and measurement
	M-02	Estimated waste avoided: derived from M-01 [OQ-05]
	M-03	Completed pickups: count of Collected
	M-04	Redistribution rate: collected ÷ published quantity
Marketplace	MK-01	Active providers (≥1 published listing)
	MK-02	Active recipients (≥1 reservation)
	MK-03	Published listings
	MK-04	Listing-to-reservation rate
	MK-05	Reservation-to-pickup rate (Collected ÷ created)
	MK-06	Repeat usage (≥2 actions in a period; window TBD)
Business hypotheses	B-01 to B-07	Paid businesses · conversion (paid ÷ active providers) · MRR · revenue per business · churn and retention (cohorts) · CAC · contribution margin
AI	A-01	Salwa usage (AI logs)
	A-02	Matchmaker usage (/api/matchmaking calls)
	A-03	AI listing completion time vs. non-AI baseline [OQ-04]
	A-04	Recommendation engagement (open/reserve events)
	A-05	AI-assisted workflow completion (completed ÷ started)

6 Target Users & Personas

Persona	Goals	Pain points	Key needs
Provider Restaurants, cafés, bakeries, hotels, caterers, commercial and wedding kitchens	Reduce waste, publish fast, reach nearby recipients, control pickup, track performance	Unpredictable demand, short shelf life, no fast liquidation, no handling visibility	Fast listing, pickup control, reservation visibility, pricing and impact insight
Individual Recipient	Find affordable or free food quickly, trust the provider, reserve and collect easily	Cannot see nearby surplus, hygiene and identity doubts, checkout friction, tight budget	Distance, price, quantity on one card, budget filter, voucher, clear pickup instructions
NGO / Institution NGOs, shelters, hostels, universities, welfare orgs	Receive surplus at scale, coordinate, keep records, get alerts, track impact	Cannot see surplus in time, cannot measure impact	Radius alerts, coordination, redistribution records, impact reports. Data model needs an NGO role [OQ-06]
Admin / Moderator	Verify providers, moderate listings, handle incidents, enforce policy, monitor analytics	Unsafe content, fraud, untraceable incidents	Review queues, audit log, suspend/restore, incident workflow, analytics

7 Core User Journeys

Persona	Flow
Provider	Sign up (role) → submit verification → Admin approves → create listing (name, quantity, unit, price/free, pickup window, location, photo, description, handling info) → optional AI Listing Assistant (editable) → submit → moderation → published → reservation notification → prepare pickup → handover → complete
Recipient	Sign up and verify → Discover → filter or AI match (budget, party size, distance) → view details → reserve and confirm → pay (cash or online) → receive voucher with unique reference → collect in window → complete → feedback
NGO / Institution	Sign up → verification → set preferences and radius → receive alerts → discover → reserve or coordinate → pick up → record redistribution
Admin	Log in → dashboard → review users and providers → moderate listings → monitor reservations → handle incidents → review analytics

8 Scope

Release	Included
MVP (P0)	Registration/login, role selection, phone/account verification · provider verification, listing moderation, incident reporting, audit trail · create/edit listing with food details, quantity, price or free, availability time, pickup info · discovery, search and filter · reservation, status, pickup voucher · cash and basic online payment · notifications, admin dashboard · “basic AI” (undefined [OQ-07])
V1	Stronger AI matching, AI listing assistant, advanced analytics, business dashboard, geo-alert optimization, improved verification, institutional workflows
V2	Third-party delivery, premium subscriptions, targeted promotion, AI business copilot, institutional contracts, sponsorships, nationwide optimization
Out of scope	Any delivery or own rider fleet · premium subscriptions and paid promotion · contracts and sponsorships · SMS/WhatsApp · permanently: AI food-safety certification, sale of raw personal data

Premium B2B package candidates (AI Listing Assistant, Business Copilot, Audience Insights, AI Reporting, Smart Matching, Food-Safety RAG Assistant, waste analytics, performance dashboard) are hypotheses to be validated; release phase unspecified **[OQ-08]**.

Scope conflicts and how this PRD resolves them

- **[OQ-09]** The draft’s FR table tags delivery V1 but scope and roadmap say V2; **this PRD uses V2**.
- **[OQ-10]** P0/P1 labels do not map to MVP/V1/V2 (“basic AI” is MVP but AI features are P1).
- **[OQ-11]** Geo-alerts are P1 but “new nearby food” is a P0 notification; basic notification is treated as P0, geo-targeting as P1.

9 Functional Requirements

Personas: PRV Provider, RCP Recipient, NGO, ADM Admin, ALL. Priorities are the draft’s labels. FR-25 to FR-27 are added because the MVP list requires them. All requirements are **must** unless stated.

ID	Requirement	Pri	Persona	Acceptance criteria (Given / When / Then)
FR-01	User registration and login	P0	ALL	Given valid details submitted then account created, login works
FR-02	Role selection at sign-up (Provider, Recipient, NGO; Admin not self-selectable)	P0	ALL	No role chosen → registration cannot complete
FR-03	Phone/account verification before reserve or publish [OQ-18]	P0	ALL	Unverified user attempts action → blocked with prompt
FR-04	Admin approves provider before publishing	P0	PRV, ADM	Pending provider submits listing → rejected with message
FR-05	Create listing: name, photo, description, quantity, unit, price/free, original price, pickup window, location, handling info	P0	PRV	Verified provider enters all required fields → can submit; missing field → blocked and identified
FR-06	Edit own listing; edits to published listings follow moderation rule [OQ-17]	P0	PRV	Owner edits → change stored and audited; other provider’s listing → denied
FR-07	Listing moderation: approve, reject, edit/flag, suspend, remove	P0	ADM	Admin approves → published; rejected → provider notified with reason
FR-08	Discovery cards: price, quantity, availability, provider, distance, trust indicators, pickup info	P0	RCP, NGO	Active listing → card shows those fields
FR-09	Search and filter: distance, price, free/paid, category, quantity, availability, provider type	P0	RCP, NGO	Filters applied → only matches; no match → empty state
FR-10	Reserve (Reserve → Confirm → Payment → Voucher)	P0	RCP, NGO	Available listing confirmed → inventory updates, unique reference generated
FR-11	Voucher with unique #RSV-XXXX reference	P0	RCP, NGO	Confirmed reservation → voucher shows reference, listing, window; references never repeat

ID	Requirement	Pri	Persona	Acceptance criteria (Given / When / Then)
FR-12	Cash at pickup and online payment where configured; payment status separate from reservation status	P0	RCP, NGO	Cash chosen → Pending until settled; online success → Paid, reservation status unchanged by payment alone
FR-13	Show reservation and payment status to both parties	P0	RCP, PRV, NGO	Status changes → both see it on next page load
FR-14	Send notifications in Section 14	P0	ALL	Trigger occurs → recipient notified on configured channel
FR-15	Incident reporting: food quality, safety, fraud, harassment, pickup issue	P0	ALL	Submitted with category → appears in Admin queue
FR-16	Audit log of Admin actions and changes to listings, reservations, verification, accounts	P0	ADM	Admin action → entry with actor, action, target, time; not UI-editable
FR-17 to 21	Salwa, AI matching, listing assistant, quality AI, geo-alerts (Section 10)	P1	per Sec. 10	See Section 10
FR-22	Advanced analytics (should)	P1	PRV, ADM	Role-relevant metrics from Section 5 shown; set TBD
FR-23	Third-party delivery (may)	V2	RCP, PRV	Deferred; criteria written when V2 approved
FR-24	Premium B2B AI package (may), hypothesis; draft says V1, scope says V2 [OQ-12]	V1	PRV	Deferred pending pilot evidence
FR-25	Admin dashboard: users, providers, listings, reservations, incidents, analytics	P0	ADM	Admin reaches all sections; non-Admin denied
FR-26	Handle cancellation and expiry of reservations and listings [OQ-15]	P0	RCP, PRV	Window passes uncollected → Expired, both notified
FR-27	Mark Collected on arrival in window with valid voucher; confirmer undefined [OQ-14]	P0	PRV, RCP	Valid reservation in window → can be collected; expired or cancelled → cannot

10 AI Features

Rules for all AI. **AI-G1** answer only from authoritative backend data, never invent listings, prices or providers · **AI-G2** never certify food as safe · **AI-G3** AI listing output must be editable before submission · **AI-G4** never override safety rules, moderation or verification · **AI-G5** log AI interactions (retention [OQ-22]) · **AI-G6** core loop must work if AI fails.

Failure handling below is derived from these rules and needs confirmation [OQ-22].

Feature	Purpose and inputs	Outputs	Guardrails	Failure handling
Salwa FR-17, P1	Conversational concierge in English, Urdu, Roman Urdu. Inputs: message, location, budget, tool results (food search, budget, distance, vendor research)	Answers referencing real listings	AI-G1 to G6; reply in user's language; no safety claims; no private addresses	Tool or model fails → say so, point to Discover; manual discovery stays usable
Smart Matchmaker FR-18, P1	Rank food by budget, party size, location, price, quantity, availability, remaining time	Ranked list with reason, e.g. "Rs 500 mein 4 logon ke liye 2.3 km away biryani best match hai"	Prices and distances from backend; only published, available listings; weights undefined [OQ-20]	No location → rank without distance and say so; nothing fits → say so, may show nearest
Listing Assistant FR-19, P1	Free text such as "20 biryani boxes, 6pm tak pickup" plus optional photo	Draft title, quantity, description, pickup window, missing fields, pickup summary	Suggestion only, fully editable; no food-safety claims; moderation still applies	Parsing fails → form usable, raw input kept
Food Quality AI FR-20, P1	Listing photo (Gemini Vision) and vendor research (Tavily)	Observations on visible condition, packaging, image quality, visual concerns, vendor trust indicators	Must not claim food is safe from an image. Always show the mandatory notice (below). Advisory only; no auto-approve	Analysis fails → moderation continues, "no AI insight available"
Pricing Suggestions P1 [OQ-10]	Remaining time, quantity, demand, location, history	Suggested price shown to provider	Provider-controlled; never changes price without provider action; tiers undefined [OQ-20]	No suggestion → provider sets price

Feature	Purpose and inputs	Outputs	Guardrails	Failure handling
Geo-Alerts FR-21, P1	New published listing, subscriber location, radius, preferences	Email (Resend) and in-app alert	Approximate distance only; published listings only	Send fails → logged, listing still publishes; retry undefined [OQ-21]

Mandatory Food Quality notice: “AI quality insights available. Final food-safety responsibility remains with the provider, platform policies and applicable requirements.”

Key acceptance criteria

- Salwa names only listings that exist in platform data and says so when nothing matches.
- Matchmaker never shows a paid listing as within budget if price exceeds it, and never ranks expired or fully reserved listings.
- The “20 biryani boxes” input yields an editable draft with quantity 20 and pickup ending 6 pm.
- Any quality result carries the mandatory notice, and a “no concerns” result still needs the normal moderation decision.
- A subscriber in radius with alerts on gets an alert; alerts off gets none.

11 Reservation, Payment & Pickup Logic

Reservation status and payment status are **independent state machines**, stored in separate fields. *Available* is treated as a listing state **[OQ-13]**. Transitions below are **proposed [OQ-14, OQ-15, OQ-16]**.

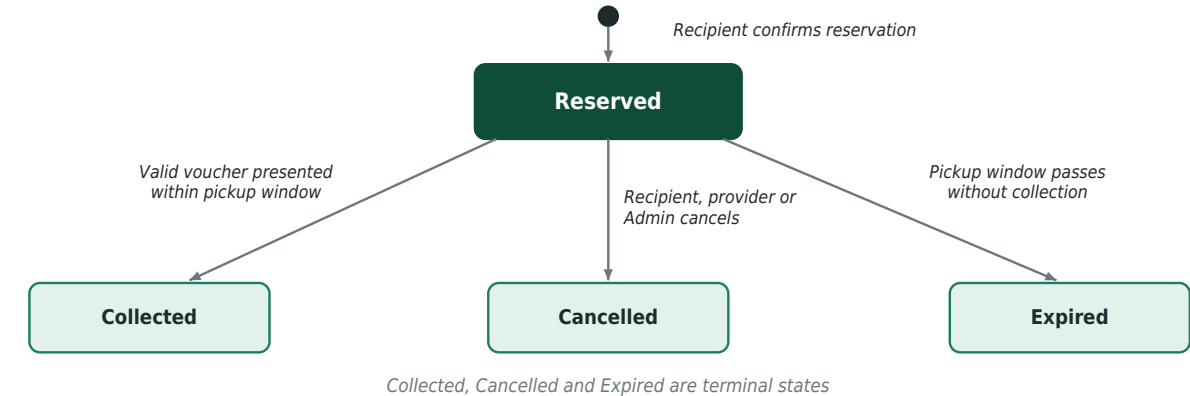
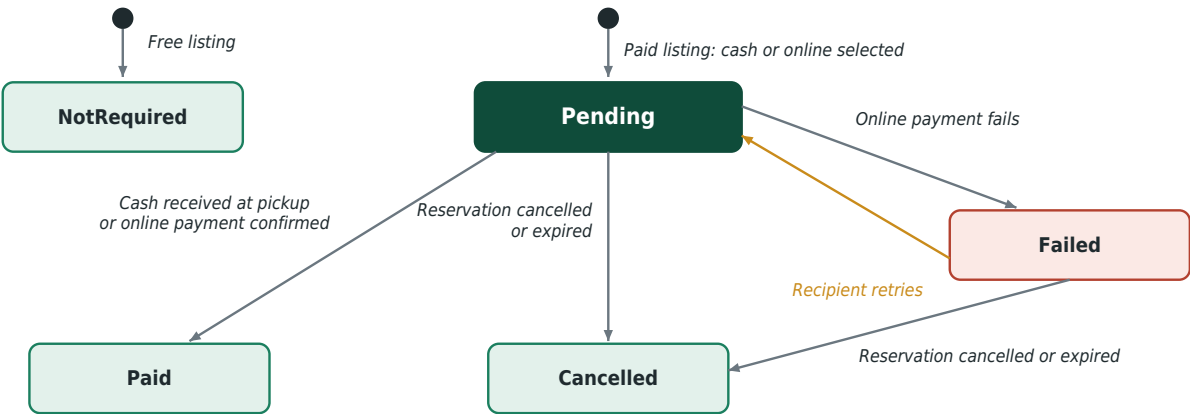


Figure 1. Reservation states (proposed)



Refunds are undefined in this version [OQ-16]
Figure 2. Payment states (proposed; refunds undefined)

Collected, Cancelled and Expired are terminal. Cancellation is by recipient, provider or Admin and must be audited; expiry is automatic.

Rule	Requirement
PAY-1 / PAY-2	Payment status stored separately; free listing is <i>NotRequired</i> .
PAY-3 / PAY-4	Cash: voucher issued at selection, <i>Pending</i> until cash received. Online: voucher only after payment confirmed.
PAY-5 / PAY-6	Payment failure must not lock a listing [OQ-16] . No charging merely for access to donated food; free vs. donated undefined [OQ-03, OQ-28] .
PU-1 to PU-3	Pickup is the only MVP fulfilment; every listing has a pickup window; recipient gets voucher and instructions.
PU-4	Exact address shown only to a recipient with a confirmed reservation [OQ-21] .
PU-5 / PU-6	Collection confirmed in system (FR-27); time-based cancellation and expiry automatic (FR-26).
PU-7	Reservation state stays consistent: Given quantity 1 and two simultaneous confirmations, then exactly one reservation succeeds and the other sees "no longer available".

12 Trust, Safety & Privacy Requirements

ID	Requirement and acceptance
TS-01	Verify providers before publishing; unverified provider cannot publish.
TS-02	Verify recipients where required [OQ-18] .
TS-03	Moderate every listing before visibility unless a documented exception applies [OQ-17] ; <i>pending</i> listings never appear in Discover.
TS-04	Listing detail shows provider, food, quantity, price/free, pickup window, handling info, status (PR-1).
TS-05 / TS-06	Incident reporting (FR-15) and audit log (FR-16).
TS-07	Suspend and restore accounts; suspended account cannot publish, reserve or access protected areas.
TS-08 / TS-09	Clear food-safety disclaimers on listing pages; safety education content, scope undefined [OQ-19] .
TS-10 / TS-11	AI never overrides safety rules or certifies food. Food categories, handling rules and local compliance reviewed before public launch.
PV-01 to PV-06	Minimize personal data (each field has a purpose); no public private addresses; role-based access via RLS; authorized analytics only; no sale of raw personal data; auditability maintained. Retention, Pakistani data-protection rules and consent flows undefined [OQ-23] .

13 Admin & Moderation Requirements

All Admin actions must be audit-logged (FR-16) and restricted to Admin accounts (ADM-07; non-Admin gets access denied).

ID	Area	Requirement and acceptance
ADM-01	Users	View, verify, suspend, restore, review roles; suspension updates status and writes audit entry.
ADM-02	Providers	Approve, reject, review documents, monitor activity [OQ-18] ; approval enables publishing, rejection notifies provider.
ADM-03	Listings	Approve, reject, edit/flag, suspend, remove; suspended listing leaves Discover.
ADM-04	Reservations	Monitor, investigate, cancel where necessary; cancellation sets <i>Cancelled</i> and notifies both parties.
ADM-05	Incidents	Handle food-quality, safety, fraud, harassment, pickup issues; severity and response targets undefined [OQ-24] .
ADM-06	Analytics	Active providers, listings, reservations, successful pickups, food redistributed, estimated waste avoided for a chosen period.

Moderation rules: *pending* listings enter a review queue; AI quality insight may be shown but is labelled advisory; the decision is made by a human or documented rule, **never by AI alone**.

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Notifications

ID	Type	Trigger	To	Pri.
N-01	Reservation confirmed	Reservation created	Recipient	P0
N-02	Reservation received	Reservation on provider’s listing	Provider	P0
N-03	Pickup reminder	Before pickup window [OQ-27]	Recipient	P0
N-04	Reservation cancelled	Status <i>Cancelled</i>	Both	P0
N-05	Listing expiring	Near end of availability [OQ-27]	Provider	P0
N-06	Pickup completed	Status <i>Collected</i>	Both	P0
N-07	Incident/update	Incident or Admin action affects user	Reporter, affected user	P0
N-08	New nearby food	Listing published in radius	Subscriber	P0 basic, P1 geo [OQ-11]

Channels: in-app and email (Resend) in MVP; SMS and WhatsApp future.

- **NT-01** send each type on trigger (delay target TBD)
- **NT-02** record type, recipient, channel and status
- **NT-03** respect user preferences
- **NT-04** a notification failure must not roll back the triggering action
- **NT-05** no private address unless the recipient has a confirmed reservation

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Data Model

Entity	Status	Key fields
Profile	Core	id, name, email, phone, role, location, verification, tier, rating, timestamps
FoodPost (food_posts)	Core	id, provider, food name, quantity, unit, price, original price, availability/expiry, location, photo, description, status, timestamps
Transaction	Core	buyer, seller, food, amount, payment method, payment status, transaction status, timestamps
Eight more	Proposed	Reservation, PickupRecord, Notification, VerificationRecord, Incident, AuditLog, FoodSubscription, AllInteractionLog. Minimal fields: id, owner/actor, reference or type, status, timestamps; details need confirmation [OQ-23]

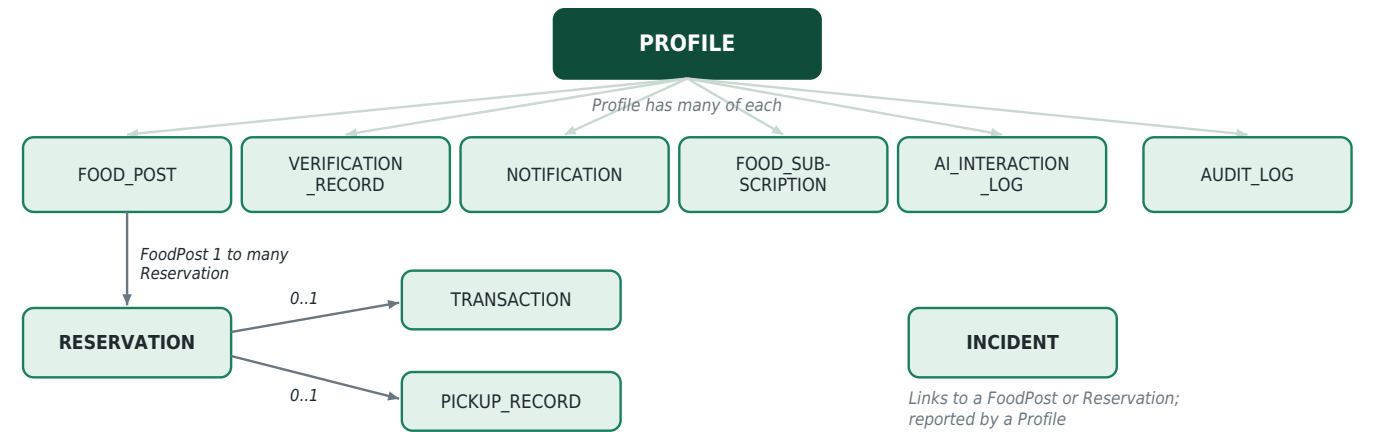


Figure 3. Entity-relationship overview (simplified; fields per table above)

Relationships: Profile 1-to-many FoodPost, Reservation, VerificationRecord, Notification, Incident, FoodSubscription, AllInteractionLog, AuditLog · FoodPost 1-to-many Reservation · Reservation 1-to-0..1 Transaction and PickupRecord · Incident links to a FoodPost or Reservation.

Open: NGO role [OQ-06], “tier” and “rating” meaning [OQ-23, OQ-26], free reservations and Transactions [OQ-16].
Proposed listing statuses: *draft, pending moderation, published, rejected, suspended, removed, expired* [OQ-17].

Data rules

- **DM-01** RLS on all personal/transactional tables · **DM-02** separate reservation and payment status
- **DM-03** append-only audit log · **DM-04** unique reservation reference · **DM-05** quantity never below zero under concurrency

16 System Architecture & API Overview

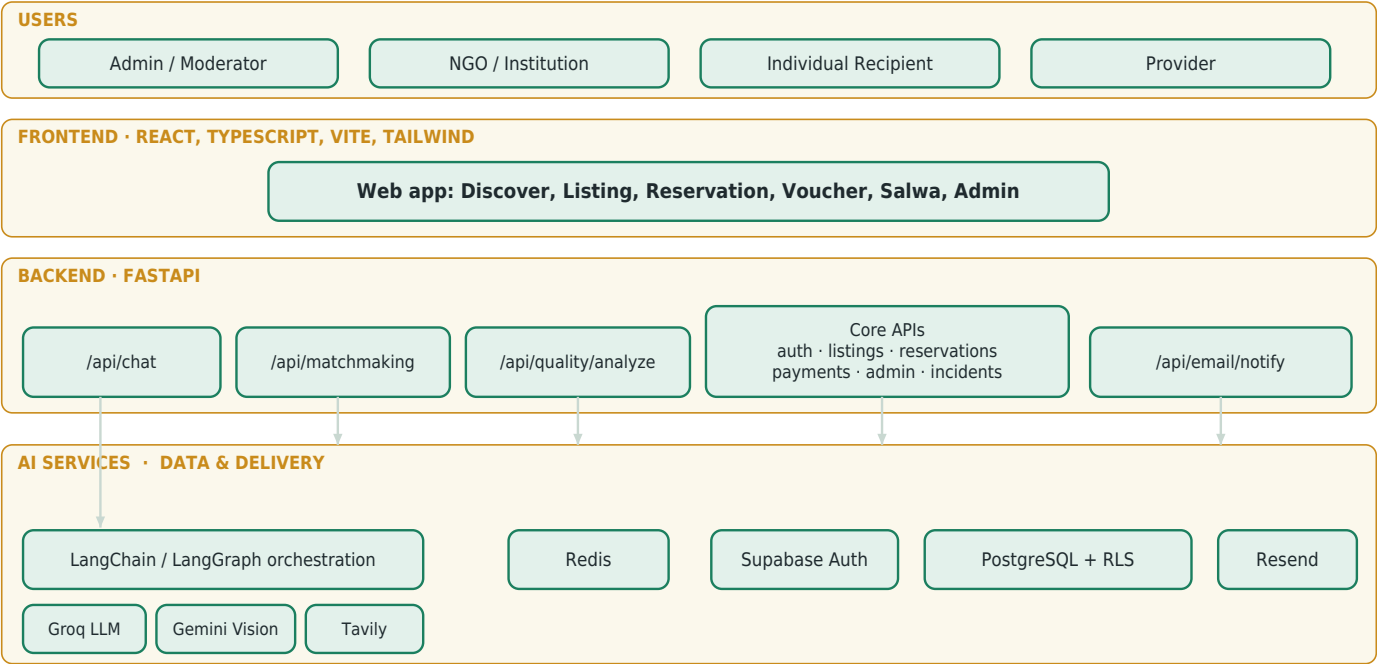


Figure 4. Logical architecture (hosting mapping unspecified [OQ-30])

API area	Purpose and rules
/api/chat	Salwa; uses backend tools, follows AI-G1 to G6.
/api/matchmaking	Budget, party-size, location ranking; published, available listings only.
/api/quality/analyze	Quality assistance; returns mandatory notice, never certifies safety.
/api/email/notify	Geographic and event emails; failure must not roll back actions.
/api/workspace	“Workspace/coordination”; purpose undefined [OQ-29].
Production areas	Authentication, listings, reservations, payments, users, providers, admin moderation, incidents, notifications; contracts undefined [OQ-29].

Architecture requirements

- **AR-01** backend is authoritative for listing, price, quantity, distance and reservation data; AI reads it through tools
- **AR-02** no secrets in the frontend (Groq, Gemini, Tavily, Resend, payment keys server-side)
- **AR-03** every route authenticates and authorizes by role · **AR-04** AI routes fail independently of core routes · **AR-05** validate all inputs

17 UX, Responsive & Accessibility Requirements

The current UI is the baseline and must be preserved (baseline artifacts not attached **[OQ-37]**). Redesign must improve clarity, hierarchy, usability, responsive behavior, consistency, accessibility and conversion. Without Product Owner approval, do **not** change: core identity, primary navigation concept, AI-focused positioning, core discovery experience, established visual language.

- **UX-01** every UI change names the focus area it serves · **UX-02** changes to protected items need approval
- **UX-03** every flow has loading, empty and error states · **UX-04** listing creation in minimal steps (target TBD **[OQ-33]**)

Responsive

Desktop, tablet and mobile supported. Mobile priority order: Discover, Food details, Reserve, Voucher, Salwa, Provider listing, Notifications. **RS-01:** Discover → Reserve → Voucher fully usable with no horizontal scroll. Browsers, devices and UI languages including Urdu/RTL are undefined **[OQ-32]**.

Accessibility (AX-01 to AX-08)

Full keyboard operation · visible focus · sufficient contrast (standard TBD) · semantic controls and labels · alt text · specific readable errors · touch-friendly targets (size TBD) · no critical information by color alone (statuses also as text).

18 Non-Functional Requirements

Numeric targets are **TBD** and must be set before the pilot **[OQ-31]**.

Area	Requirements
Performance	PERF-01 fast core pages on typical Pakistani mobile networks · PERF-02 API latency target · PERF-03 AI latency target; slow AI must not block core flows
Reliability	REL-01 one winner for concurrent reservations · REL-02 AI, notification or email failure must not corrupt core data · REL-03 payment and reservation status stay separate and consistent
Security	SEC-01 Supabase Auth · SEC-02 role authorization · SEC-03 RLS on personal/transactional tables · SEC-04 secured API · SEC-05 input validation · SEC-06 rate limiting, limits undefined [OQ-36] · SEC-07 secrets outside source code
Scalability	SCL-01 support growth from pilot geography to Pakistan-wide use; capacity and pilot geography undefined [OQ-31, OQ-34]
Observability	OBS-01 to OBS-06 monitor API errors, AI errors, reservation failures, payment failures, notification failures, latency; each failure recorded and visible to engineering. Tooling and thresholds undefined [OQ-39]

19 Risks & Mitigations

Mitigations come from the draft. Likelihood (L), impact (I) and owner are **proposed** and need confirmation **[OQ-33]** (H high, M medium, L low).

ID	Risk	L	I	Mitigation	Owner
R-01	Food safety	M	H	Rules, moderation, compliance review; AI never certifies	Compliance
R-02	AI hallucination	M	H	Backend-authoritative data	Engineering
R-03	Provider adoption	M	H	Simple listing workflow; pilot validation	Product Owner
R-04	Pickup failure	M	M	Defined windows and reminders	Product Owner
R-05	Fraud	M	H	Verification and incident reporting	Admin
R-06	Privacy	L	H	Controlled location and data sharing	Engineering
R-07	Low retention	M	H	Measure repeat usage (MK-06)	Product Owner
R-08	Delivery complexity	L	M	Defer to V2 until pickup validated	Product Owner
R-09	AI overreliance	M	M	Human and rule authority	Product Owner

ID	Risk	L	I	Mitigation	Owner
R-10	Overbuilding	M	M	MVP-first roadmap	Product Owner

20 Validation Plan

All items are hypotheses to be validated; success thresholds are TBD and must be set before the pilot [OQ-35].

Hypothesis	Experiments and data
H-01 Businesses will pay for premium AI	EXP-04 pricing experiments (B-02, B-04), EXP-01 provider interviews
H-02 Providers will repeatedly publish surplus	EXP-03 controlled pilot (MK-01, MK-03, MK-06), EXP-01
H-03 Recipients will repeatedly use the platform	EXP-03, EXP-02 recipient testing (Discover → Pickup), EXP-06 pickup completion analysis
H-04 Users will accept the payment workflow	EXP-05 transaction analysis, EXP-02
H-05 Delivery will have viable economics	Not tested in MVP; evaluate in V2
H-06 Institutional contracts will be valuable	NGO/institution pilot and interviews (B-01, B-03)

EXP-07 safety and incident monitoring runs throughout. **Decision rule:** Phase 8 (Scale) does not start until the evidence in Section 21 exists; “enough evidence” thresholds are open [OQ-35]. Pilot size, geography and duration are undefined [OQ-34].

21 Roadmap & Phasing

Scope is from the draft; entry/exit criteria for Phases 1–7 are **proposed**; no dates were provided; phase-to-release mapping is open [OQ-10].

#	Phase	Scope	Exit criteria (proposed)
1	Foundation	Finalize PRD, roles, data model, safety rules, analytics	PRD approved; open questions resolved or accepted
2	Core Marketplace	Auth, listing, discovery, reservation, pickup, status	FR-01 to 03, 05, 06, 08 to 11, 13, 26, 27 pass; consistent under concurrency
3	Trust Layer	Verification, moderation, incidents, audit	FR-04, 07, 15, 16, 25 pass; unverified/unmoderated cannot reach Discover
4	AI Layer	Salwa, matching, listing assistant, quality AI	FR-17 to 20 pass; AI-G1 to G6 verified; core loop works with AI off
5	Payments & Notifications	Cash, online payment, email/in-app, geo-alerts	FR-12, 14, 21 pass; payment and reservation states independent; provider confirmed [OQ-16]
6	Business Layer	Provider dashboard, reporting, analytics, premium	FR-22 passes; premium labelled hypothesis
7	Validation	Pilot, interviews, feedback, pricing tests	EXP-01 to 07 complete and compared to thresholds
8	Scale	Pakistan-wide expansion	Entry only after evidence of repeat usage, reliable pickup, safety, provider retention, sustainable economics

22 Definition of Done & Launch Criteria

Definition of Done (DOD-01 to 11): acceptance criteria pass · critical defects resolved · security and privacy requirements pass · mobile experience works · error states exist · analytics implemented · user-facing copy complete · documentation exists · QA complete · Product Owner approves.

Launch criteria

Public Pakistan-wide launch; pilot gating unclear [OQ-34].

Area	IDs	Criteria
Product	LC-01 to 03	Core workflows stable, reservation state reliable, pickup process tested

Area	IDs	Criteria
Safety	LC-04 to 07	Food categories reviewed, handling rules defined, incident process operational, local compliance reviewed [OQ-38]
Business	LC-08 to 10	Provider demand, recipient demand and monetization assumptions validated (H-01, H-02, H-03, H-06)
Technical	LC-11 to 15	Security, performance, API and mobile/browser testing complete, monitoring in place

23 Open Questions & Assumptions

Group	Open questions
Product and scope	OQ-01 named owner and stakeholders · OQ-02 market size and source · OQ-03/28 what NG-07 prohibits; “free” vs “donated” · OQ-04 KPI targets, repeat window, non-AI listing baseline · OQ-05 serving conversion and waste estimate · OQ-06 NGO role in data model · OQ-07 what “basic AI” includes · OQ-08 release of each premium item · OQ-09 delivery V1 or V2 · OQ-10 P-label to release mapping · OQ-11 geo-alerts P0 or P1 · OQ-12 FR-24 V1 or V2
Reservation, payment, listings	OQ-13 whole vs partial quantity, inventory release · OQ-14 who confirms collection, voucher validation · OQ-15 hold time, cut-off, grace, no-show rules · OQ-16 payment states, refunds, cash settlement, Pakistani payment provider, free-listing transactions · OQ-17 listing statuses, moderation scope, re-moderation on edit · OQ-25 NGO redistribution records, bulk reservations · OQ-26 feedback and ratings
Trust, privacy, AI	OQ-18 phone and provider verification, recipient verification triggers · OQ-19 food categories, handling fields, safety education · OQ-20 matchmaker weights, markdown tiers · OQ-21 location precision, address reveal, alert radius, frequency, retry · OQ-22 AI fallback, log retention, provider fallback · OQ-23 retention, consent, Pakistani data-protection law, “tier”, entity fields · OQ-24 incident severity and response targets · OQ-38 local food-safety regulation and reviewer
Delivery and operations	OQ-27 reminder and expiry lead times, channels, preferences · OQ-29 /api/workspace purpose and API contracts · OQ-30 Redis purpose, Vercel/Render mapping · OQ-31 numeric performance, capacity targets · OQ-32 browsers, devices, languages, accessibility standard, touch size · OQ-33 risk ratings and owners, listing step count · OQ-34 pilot geography, size, duration, gating · OQ-35 experiment thresholds · OQ-36 rate limits, auth and session policy · OQ-37 UI baseline artifacts · OQ-39 observability tooling and thresholds

Assumptions (all need confirmation)

ID	Assumption
A-01	Delivery is V2; the draft’s “V1” tag is an error
A-02	FR-25 to 27 added because the MVP list requires them
A-03	<i>Available</i> is a listing state; the other four are reservation states
A-04	State transitions in Section 11 are proposed
A-05	Listing statuses and proposed entity fields are proposed
A-06	NGO role added
A-07	AI failure handling derived from AI-G1 to G6
A-08	Risk ratings and phase criteria proposed
A-09	All KPI and validation thresholds are TBD

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Glossary

Term	Meaning
Admin/Moderator	staff who verify, moderate and handle incidents
Audit log	append-only action record
Collected	terminal status after valid pickup
Discover	the browse-and-filter interface
Food rescue	moving edible surplus to those who can use it
Geo-alert	location-based new-listing notification
Hypothesis	to be validated claim without pilot evidence
Listing	a provider's surplus offer (food_posts)
Matchmaker	AI ranking by budget, party size, distance
NGO/Institution	organization receiving surplus
P0/P1	the draft's priority labels
Pickup window	time range for collection
Provider	food business publishing surplus
Recipient	individual or organization reserving food
Reservation	confirmed claim with #RSV-XXXX reference
RLS	Row Level Security
Salwa	the AI assistant
Surplus	edible food a provider cannot sell or use
Trust layer	verification, moderation, incidents, audit, disclaimers
Voucher	confirmation shown at pickup

✓

PRD Quality Checklist

Check	Status	Note
All requirements testable	Partially met	Given/When/Then or verifiable rule throughout; NFR and KPI targets TBD [OQ-04, OQ-31] ; FR-23 and FR-24 deferred by design
Hypotheses labelled	Met	Sections 2, 5, 8, 20 and FR-24; thresholds TBD [OQ-35]
AI safety rules included	Met	AI-G1 to G6, mandatory notice, never certify from an image, TS-10, human moderation
Open questions listed	Met	39 questions and 9 assumptions (Section 23)
Preserved decisions	Met	Pickup-first MVP, delivery in V2, B2B-first hybrid, current UI baseline, AI safety rule
No invented content	Met with flags	Proposed items tagged and listed in assumptions

Before approval, resolve first: OQ-09/10 (scope labels) · OQ-13 to 17 (reservation, payment, listing status) · OQ-18 (verification) · OQ-31/35 (targets and thresholds).